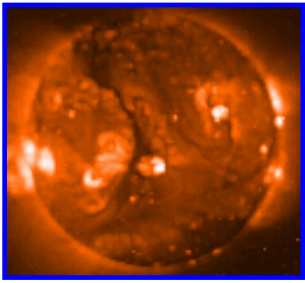


Sun Fact Sheet



Sun/Earth Comparison

Bulk parameters

	Sun	Earth	Ratio (Sun/Earth)
Mass (10^{24} kg)	1,989,100.	5.9736	332,950.
GM ($\times 10^6$ km ³ /s ²)	132,712.	0.3986	332,950.
Volume (10^{12} km ³)	1,412,000.	1.083	1,304,000.
Volumetric mean radius (km)	696,000.	6371.	109.2
Mean density (kg/m ³)	1408.	5520.	0.255
Surface gravity (eq.) (m/s ²)	274.0	9.78	28.0
Escape velocity (km/s)	617.7	11.2	55.2
Ellipticity	0.00005	0.0034	0.015
Moment of inertia (I/MR ²)	0.059	0.3308	0.178
Visual magnitude V(1,0)	-26.74	-3.86	-
Absolute magnitude	+4.83		
Luminosity (10^{24} J/s)	384.6		
Mass conversion rate (10^6 kg/s)	4300.		
Mean energy production (10^{-3} J/kg)	0.1937		
Surface emission (10^6 J/m ² s)	63.29		
Spectral type	G2 V		

Model values at center of Sun:

Central pressure: 2.477×10^{11} bar

Central temperature: 1.571×10^7 K

Central density: 1.622×10^5 kg/m³

Rotational and Orbital parameters

	Sun	Earth	Ratio (Sun/Earth)
Sidereal rotation period (hrs)*	609.12	23.9345	25.449
Obliquity to ecliptic (deg.)	7.25	23.45	0.309
Speed relative to nearby stars (km/s)	19.4		

*This is the adopted period at 16 deg. latitude - the actual rotation rate varies with latitude L as:
 ($14.37 - 2.33 \sin^2 L - 1.56 \sin^4 L$) deg/day

Sun Observational Parameters

Apparent diameter from Earth	
At 1 A.U.(seconds of arc)	1919.
Maximum (seconds of arc)	1952.
Minimum (seconds of arc)	1887.
Distance from Earth	
Mean (10^6 km)	149.6
Minimum (10^6 km)	147.1
Maximum (10^6 km)	152.1

Solar Magnetic Field

Typical magnetic field strengths for various parts of the Sun

Polar Field: 1 - 2 Gauss
 Sunspots: 3000 Gauss
 Prominences: 10 - 100 Gauss
 Chromospheric plages: 200 Gauss
 Bright chromospheric network: 25 Gauss
 Ephemeral (unipolar) active regions: 20 Gauss

Solar Atmosphere

Surface Gas Pressure (top of photosphere): 0.868 mb
 Effective temperature: 5778 K
 Temperature at bottom of photosphere: 6600 K
 Temperature at top of photosphere: 4400 K
 Temperature at top of chromosphere: ~30,000 K
 Photosphere thickness: ~400 km
 Chromosphere thickness: ~2500 km

Sun Spot Cycle: 11.4 yr.

Photosphere Composition:

Major elements: H - 90.965%, He - 8.889%

Minor elements (ppm): O - 774, C - 330, Ne - 112, N - 102
Fe - 43, Mg - 35, Si - 32, S - 15

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